UNIVERSITY OF CHICAGO
BOOTH SCHOOL OF BUSINESS

B30801 Managerial Accounting

ZPhone Contribution Margins.

ZP Company is making plans for production of a new product, the ZPhone, that it hopes will be a game-changing mobile device. There are two variants of this device, ZPhone-K and the more advanced ZPhone-X. Relevant pricing and cost data are as follows:

	Model K	Model X
Retail price	$750	$1,000
Direct labor	1.5 hours @ $18	2 hours @ $18
Direct materials:		
Battery cells	1 @ $10	2 @ $10
Processor	$45	$45
Touch screen	$50	$75
Other materials	$150	$170

Monthly fixed costs of running the plant are $3.0M, which provides capacity of up to 11,000 units/month; these costs increase to $6.5M/month if production exceeds this threshold (which requires a second shift) to a maximum capacity of 20,000 units/month.

Required:

1. The company initially plans to produce only Model K and limit production to no more than 11,000 units. Compute the monthly break-even point. How many units should the company produce if it can sell all the units it makes?

2. After its first two months of operations, Model K has proved very popular with customers, selling nearly 11,000 units/month, and the company decides to expand production by adding a second shift. Compute: (i) monthly break-even point(s), and (ii) how many units it should produce to maximize profit.

3. Having achieved this success, the company now brings Model X to market. How many units of each product should it produce, assuming it initially decides to again limit total production to no more than 11,000 total units?

4. Due to supply chain issues, the company can only obtain 10,000 battery cells per month from its suppliers. How many units of each model should it produce, and what total contribution margin would result?

5. How would total contribution margin change if the company was able to obtain another 500 battery cells each month?

6. How does your answer to 4. change if *in addition* to the battery cell constraint, the company can only obtain 8,000 processors each month? (Note: you will not see a problem like this -- with more than one constraint -- on the final exam.)